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Wood bundling and bagging system

Abstract

A system and method for bundling wood includes a bin portion collecting logs that drop off a conveyor platform in a transverse orientation relative to a direction of travel. A first sensor detects each log as it drops, forming a stack, and a control module maintains a running count of the logs in the stack. A hatch automatically opens and simultaneously drops a stack of logs therethrough based on the running count, e.g., based on the running count equaling a preselected number. A second sensor above the bin portion measures a distance from the second sensor to the stack. The hatch automatically opens to drop the stack therethrough to a lower bin in response to the running count reaching the preselected number of logs and/or the distance reaching a predetermined distance. A third sensor positioned adjacent a lower hatch in the lower bin automatically closes the hatch when blocked by the stack.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of and priority to U.S. Provisional Application Ser. No. 63/410,911 entitled “WOOD BUNDLING SYSTEM,” filed Sep. 28, 2022, the entirety of which is hereby incorporated herein by reference thereto.

FIELD OF DISCLOSURE

(1) The present disclosure relates generally to a wood bundling system and, in particular, to an automated bundling system for wood, for example, firewood logs and to a bundling system including a bagging portion.

BACKGROUND

(2) There are systems known for automated bundling of firewood logs, but they are complex systems which include complicated one-by-one capturing and aligning of each log on a hatch before adding it to a stack.

(3) Accordingly, there is still a need for a more efficient system and method for automated bundling of wood in aligned stacks for bagging, as well as for bagging the stacks of wood in an efficient manner.

SUMMARY

(4) Features of the disclosure will become apparent from the following detailed description considered in conjunction with the accompanying drawings. It is to be understood, however, that the drawings are designed as an illustration only and not as a definition of the limits of this disclosure.

(5) The present disclosure is directed to a system for bundling and bagging wood as shown and described herein. The present disclosure is also directed to a method of bundling wood as shown and described herein.

(6) The present disclosure is directed to a system for bundling wood, for example, that includes a bin-adjacent conveyor platform configured to transport logs in a transverse orientation relative to a direction of travel of the bin-adjacent conveyor platform, such that a longitudinal axis of each of the logs on the bin-adjacent conveyor platform is in a transverse orientation relative to the direction of travel. The system also includes a bin portion, wherein an end of the bin-adjacent conveyor platform is positioned above the bin portion, such that each log sequentially drops off the end of the bin-adjacent conveyor platform in the transverse orientation into the bin portion.

(7) The system may also include a first sensor positioned before the end of the bin-adjacent conveyor platform; a lower hatch formed in a bottom of the bin portion; and a control module, operatively connected to the first sensor. The first sensor is configured to detect each log as it passes the first sensor in the transverse orientation and before it sequentially drops off the end of the bin-adjacent conveyor platform. The control module, in embodiments, includes a processor, and is configured to maintain a running count of the logs detected by the first sensor, and wherein the lower hatch is configured to hingedly open and simultaneously drop a stack of logs therethrough based on the running count.

(8) In embodiments, the bin portion includes an upper bin including an upper hatch formed in a bottom of the upper bin and an upper actuator operatively connected to the control module and to the upper hatch. The bin portion may further include a lower bin positioned below the upper bin, and includes the lower hatch. The upper bin is configured to receive each log detected by the first sensor and sequentially dropped off the end of the bin-adjacent conveyor platform to form the stack of logs, and wherein the control module is operatively connected to the upper actuator and is configured, via the upper actuator, to automatically open the upper hatch to drop the stack of logs into the lower bin based on the running count of logs.

(9) In embodiments, the control module is configured, via the upper actuator, to automatically open the upper hatch to drop the stack of logs into the lower bin in response to the running count of logs equaling a preselected number of logs.

(10) In additional embodiments, the system further includes a second sensor positioned above the bin portion and past an end of the bin-adjacent conveyor platform, wherein the second sensor is operatively connected to and in communication with the control module and is configured to measure a distance from the second sensor to an uppermost log of the stack of logs in the upper bin, or to the upper hatch in an absence of logs in the upper bin. In embodiments, the distance is communicated to the control module, and the control module is configured, via the upper actuator, to automatically open the upper hatch to drop the stack of logs into the lower bin in response to one of the running count of logs reaching the preselected number of logs and the distance from the second sensor and the uppermost log reaching a predetermined distance.

(11) Embodiments of the system may further include a third sensor positioned and configured to

detect a presence of logs on the lower hatch in the lower bin. The third sensor is operatively connected to and in communication with the control module. The control module may be configured, via the upper actuator, in embodiments, to automatically close the upper hatch in response to the third sensor detecting the logs in the lower bin corresponding to the stack of logs dropped into the lower bin from the upper bin.

(12) The control module may be further configured to reset the running count to zero in response to the automatic closing of the upper hatch.

(13) In further embodiments, the control module may be further configured to stop the bin-adjacent conveyor platform and then automatically open the upper hatch to drop the stack of logs in response to one of the running count of logs reaching the preselected number of logs and the distance from the second sensor and the uppermost log reaching a predetermined distance.

(14) The control module may also be configured, in embodiments, to restart the bin-adjacent conveyor platform in response to the automatic closing of the upper hatch.

(15) Embodiments of the system may further include a user control panel operatively connected to the control module and a display screen operatively connected to and associated therewith. The display screen may include at least one of a graphical indicator of the status of the upper bin and a display counter that continuously displays the number of logs in the upper bin based on the running count of logs, wherein the control module is configured to continuously update the at least one of the graphical indicator and the display counter to display the number of logs collected in the upper bin and to indicate, in response to the automatic closing of the upper hatch, an empty status of the upper bin.

(16) The system may further include, in embodiments, a user control panel operatively connected to the control module and a display screen operatively connected to and associated therewith, the display screen including a lower bin status indicator configured to display a status of the lower bin as empty or full based on the third sensor, wherein the status switches to full in response to the third sensor detecting the logs in the lower bin corresponding to the stack of logs dropped into the lower bin from the upper bin.

(17) In embodiments, the system includes a user control panel operatively connected to the control module, and a bagging portion adjacently below the lower bin. The user control panel may also include a user operable switch operatively connected to the lower actuator and configured to allow a user to open and close the lower hatch in use, via the user operable switch, to drop the stack of logs therethrough and into the bagging portion.

(18) The system, in embodiments, may further include a lower actuator operatively connected to the control module and to the lower hatch, and a bagging portion positioned adjacently below the lower bin. The control module, in embodiments, may be configured to automatically open the lower hatch, via the lower actuator, and release the stack of logs into the bagging portion in response to the third sensor detecting the presence of logs in the lower bin corresponding to the stack of logs dropped from the upper bin.

(19) In further embodiments, the system may further include a main conveyor device onto which the logs are fed in a random orientation from a log hopper; a slotted conveyor platform positioned to transport the logs that exit from the main conveyor device; and a transitional conveyor belt positioned to move and transfer the logs that exit the slotted conveyor platform to the bin-adjacent conveyor platform in the longitudinal orientation. The slotted conveyor platform, in embodiments, includes a plurality of flat plates spaced apart on each of a plurality of rotatable axles, the plurality of rotatable axles being parallel to each other and spaced apart along the direction of travel of the slotted conveyor platform, wherein the plurality of rotatable axes and the plurality of flat plates are positioned and configured to urge each of the logs in a longitudinal orientation with the longitudinal axis aligned with the direction of travel of the slotted conveyor platform.

(20) The bin-adjacent conveyor platform may intersect the transitional conveyor belt perpendicular to the direction of travel of the transitional conveyor belt, such that the logs exiting the transitional

conveyor belt are oriented in the transverse orientation relative to the direction of travel of the bin-adjacent conveyor platform.

(21) The slotted conveyor platform, in embodiments, is positioned at a height above an input end of the transitional conveyor belt.

(22) The system may further include a slide operatively connected between an exit end of the slotted conveyor platform and the input end of the transitional conveyor belt, wherein the slide is concave in cross-section transverse to the direction of travel, and wherein the direction of travel of the slide is curved and oriented to direct the logs in a downward and curved path and onto the input end of the transitional conveyor belt.

(23) In additional embodiments, the main conveyor device is positioned parallel to the transitional conveyor belt, with the direction of travel of the transitional conveyor belt being parallel to but opposite that of the main conveyor device.

(24) The system may further include a rotating arm connected to an outward extension of one axle of the plurality of rotatable axles proximate the slide and configured to operatively strike a portion of the slide as it rotates with the one axle, such that the rotating arm shakes the slide to urge the logs downward along the slide and to agitate the logs to loosen debris therefrom. The plurality of blades is also preferably configured to agitate the logs as they are urged along the direction of travel.

(25) The present disclosure is also directed to a method of bundling wood for bagging, the method including: receiving into an upper bin, logs dropped sequentially off an end of a bin-adjacent conveyor platform to form a stack of logs in the upper bin, the upper bin including an upper hatch formed in its bottom floor; providing a first sensor positioned before the end of the bin-adjacent conveyor platform, a second sensor positioned above the upper bin, an upper actuator configured to open and close the upper hatch, and a control module operatively connected to, and in communication with, the first sensor, the second sensor, the upper actuator, and the bin-adjacent conveyor platform; detecting each log, via the first sensor, as it passes the first sensor and before it sequentially drops off the end of the bin-adjacent conveyor platform; maintaining, via the control module, a running count of the logs detected, via the first sensor, and dropped into the upper bin; measuring a distance, via the second sensor, from the second sensor to an uppermost log of the stack of logs in the upper bin, or to the upper hatch in an absence of logs in the upper bin; communicating the distance, via the second sensor, to the control module; and automatically opening the upper hatch, via the upper actuator in communication with the control module, in response to one of the running count of logs reaching a preselected number of logs, and the distance from the second sensor and the uppermost log reaching a predetermined distance, thereby dropping the stack of logs into a lower bin.

(26) In embodiments, the method may further include stopping, via the control module, the bin-adjacent conveyor platform, in response to the one of the running count of logs reaching a preselected number of logs, and the distance from the second sensor and the uppermost log reaching a predetermined distance, and prior to the automatically opening the upper hatch.

(27) The method may further include providing a third sensor positioned and configured to detect a presence of logs in the lower bin; detecting the stack of logs, via the third sensor, in the lower bin; automatically closing, via the upper actuator, the upper hatch in response to the detecting the stack of logs; and restarting the bin-adjacent conveyor platform, via the control module, and resetting the running count to zero, via the control module, in response to the automatically closing the upper hatch.

(28) A lower hatch is preferably formed in a bottom of the lower bin, and a lower actuator is provided, which is configured to open and close the lower hatch. The method may further include opening and closing the lower hatch, via the lower actuator, to empty the lower bin, after the restarting the bin-adjacent conveyor platform step.

(29) The receiving into the upper bin step may further include, in embodiments, receiving each log

dropped sequentially off the end of the bin-adjacent conveyor platform in a transverse orientation, a longitudinal axis of each log being transverse to a direction of travel of the bin-adjacent conveyor platform.

(30) The present disclosure is also directed to a vertical unit for bundling a stack of wood received from a conveyor platform. The vertical unit includes an upper bin, and a lower bin positioned below the upper bin. The upper bin is positionable to receive logs dropped from the conveyor platform, and includes an upper hatch formed in its bottom floor. The lower bin includes a lower hatch formed in a bottom floor of the lower bin.

(31) The vertical unit may also include a first sensor configured to be positioned above an end of the conveyor platform, and further configured to detect each log as it passes the first sensor; and a control module operatively connected to the first sensor, the control module including a processor, and configured to maintain a running count of the logs detected by the first sensor. The vertical unit also includes an upper actuator operatively connected to the control module and to the upper hatch; and a second sensor positioned above the bin portion, wherein the second sensor is operatively connected to and in communication with the control module and is configured to measure a distance from the second sensor to an uppermost log of the stack of logs in the upper bin, or to the upper hatch in an absence of logs in the upper bin, and to communicate the distance to the control module. The control module is configured, via the control module, to automatically open the upper hatch, via the upper actuator, to drop the stack of logs into the lower bin in response to one of the running count of logs reaching a preselected number of logs and the distance from the second sensor and the uppermost log reaching a predetermined distance.

(32) The vertical unit, in embodiments, further includes a third sensor positioned and configured to detect a presence of logs on the lower hatch in the lower bin, the third sensor being operatively connected to and in communication with the control module. The control module is configured, via the upper actuator, in embodiments, to automatically close the upper hatch in response to the third sensor detecting the stack of logs dropped into the lower bin from the upper bin, and, in embodiments, to reset the running count to zero in response to the automatic closing of the upper hatch.

(33) In addition to the above aspects of the present disclosure, additional aspects, objects, features and advantages will be apparent from the embodiments presented in the following description and in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings constitute a part of this disclosure and include examples, which may be implemented in various forms. It is to be understood that in some instances, various aspects of the disclosure may be shown exaggerated or enlarged to facilitate understanding. The teaching of the disclosure can be readily understood by considering the detailed description in conjunction with the accompanying drawings, which are briefly described below.

(2) FIG. 1 is a pictorial, perspective, representation of an embodiment of a system of the present disclosure.

(3) FIG. 2A is a top view of an embodiment of FIG. 1.

(4) FIG. 2B is a perspective top view of a portion of an embodiment of the system, with a magnified view of a slotted conveyor platform configured.

(5) FIG. 2C is a perspective bottom, side view of a portion of the embodiment of FIG. 2B, showing the slide.

(6) FIG. 3 is a side view of an embodiment of a vertical unit of the system of the present disclosure, which includes a bin portion for receiving stacks of logs and a lower open portion configured for

bagging the stacks.

(7) FIG. 4 is a front view of the vertical unit of FIG. 3.

(8) FIG. 5A represents a side view of a bin portion of the vertical unit.

(9) FIG. 5B represents a front view of a bin portion of the vertical unit.

(10) FIG. 6A is a schematic representation of an embodiment of a control module and its interaction with other portions of the system of the present disclosure.

(11) FIG. 6B is a schematic representation of an embodiment of a user control panel of the present disclosure.

(12) FIG. 7 is a flow chart representation of embodiments of a method of the present disclosure.

(13) The various aspects of the present disclosure mentioned above are described in further detail with reference to the aforementioned figures and the following detailed description of exemplary embodiments.

DETAILED DESCRIPTION

(14) The following sections describe exemplary embodiments of the system and components of the system of the present disclosure and methods associated therewith. It should be apparent to those skilled in the art that the described embodiments of the present disclosure provided herein are illustrative only and not limiting, having been presented by way of example only. All features disclosed in this description may be replaced by alternative features serving the same or similar purpose, unless expressly stated otherwise. Therefore, numerous other embodiments of the modifications thereof are contemplated as falling within the scope of the present disclosure and of the system, its components, and methods associated therewith as defined herein and equivalents thereto.

(15) Particular illustrative embodiments of the present disclosure are described hereinbelow with reference to the accompanying drawings; however, the disclosed embodiments are merely examples of the disclosure, which may be embodied in various forms. It should be apparent to those skilled in the art that the described embodiments provided herein are illustrative only and not limiting, having been presented by way of example only. All features disclosed in this description may be replaced by alternative features serving the same or similar purpose, unless expressly stated otherwise. Therefore, numerous other embodiments of the modifications thereof are contemplated as falling within the scope of the present disclosure of a wood bundling and bagging system and components and methods associated therewith as defined herein and equivalents thereto. Well-known functions or constructions and repetitive matter are not described in detail to avoid obscuring the present disclosure in unnecessary or redundant detail. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting. In this description, as well as in the drawings, like-referenced numbers represent elements which may perform the same, similar, or equivalent functions.

(16) Throughout the description, where items are described as having, including, or comprising one or more specific components or features, or where methods are described as having, including, or comprising one or more specific steps, it is contemplated that, additionally, there are items of the present disclosure that consist essentially of, or consist of, the one or more recited components or features, and that there are methods according to the present disclosure that consist essentially of, or consist of, the one or more recited processing steps.

(17) The present disclosure is directed to an automated wood bundling system, for bundling firewood logs, for example. The present disclosure is also directed to vertical unit configured to perform the automatic bundling and bagging of stacks of logs received from the conveyor system, which may be incorporated as a unit into any conveyor system for bundling and bagging wood.

(18) The term “conveyor platform” as used in the present disclosure refers to the moving platform of a mechanical transport apparatus for conveying articles positioned on the conveyor platform from one location to the other via the mechanical transport apparatus. A mechanical transport apparatus is also referred to herein as a “conveyor” and the system of multiple conveyors of the

present disclosure is referred to herein as a conveyor system, or conveyor portion of the wood bundling system. The conveyor platform may be, for example, but is not limited to, the moving platform of a conventional conveyor belt system, or any other suitable mechanical transport apparatus, or conveyor apparatus.

(19) Referring to FIG. 1, an embodiment of a system **10** of the present disclosure includes a main conveyor platform **12**, onto which logs **15** are fed from a log hopper **14**, in a randomly oriented and unaligned manner, onto a first, or loading, end **16** of the main conveyor platform **12**. In embodiments, referring also to FIG. 2B, the system may further include a slotted conveyor platform **20**, which is formed from a plurality of rotating axles **21** spaced apart along the direction of travel **24** and aligned parallel to each other, the open spacing between the rotating axles **21** forming slots **19** therebetween. Each one of the rotating axles **21** includes a plurality of flat plates **23**, also referred to herein as blades **23**, each of which is fixed and centered on, and rotate with, the rotating axles, and which may be of rectangular, or square shape, positioned equidistant from each other along each one of the rotating axles **21**. As shown in FIG. 2B, in embodiments, the transverse position of each of the blades **23** along one of the rotating axles is staggered relative to the blades **23** on an adjacent rotating axle **21'**. The rotation of the rotating axles **21** and of the blades **23** therewith, moves the logs **15** along the slotted conveyor platform **20**, while the shape and positioning of the blades **23** both aligns the longitudinal axis **22** of the logs **15** in the direction of travel **24**, and agitates the logs **15** so that dirt and debris is shaken off of the logs **15**, and falls through the slots **19**.

(20) Referring also to FIGS. 2A and 2B, the logs **15** are translated along the main conveyor platform **12**, which is, in embodiments, inclined upward from the first end **16** to a second end **18**, from which the logs **15** are dropped onto the slotted conveyor platform **20**, which orients the logs **15** in the direction of travel **24** on the slotted conveyor platform **20**. In embodiments, the slotted conveyor platform **20** is substantially horizontal.

(21) Referring to FIGS. 1, 2A and 2B, in embodiments, the system may also include a slide **28** positioned at an exit end **26** of the slotted conveyor platform **20**, which may have a concave shape in cross-section transverse to the direction of travel **24**. In FIG. 1, only the ribs of the slide **28**, the structure of which is best represented in FIG. 2C, are shown so that the logs **15** and their direction of travel within the slide **28** are visible in FIG. 1. The logs **15** exit the slotted conveyor platform onto the slide **28**, which is, in embodiments, curved and oriented to redirect the logs **15** in a circular path downward and underneath the slotted conveyor platform **20** (see also FIG. 2C). From a lower end **30** (see FIGS. 2A and 2C) of the slide **28**, referring also to FIG. 1, the logs **15**, in embodiments, are loaded onto another conveyor platform **32**, referred to as a transitional conveyor platform **32**, which is substantially parallel to, and preferably less than a horizontal length of, the main conveyor platform **12**. The logs **15** are still aligned along the direction of travel **24** with the direction of travel **24** being parallel to, and, in embodiments, in the opposite direction to, that of the main conveyor platform **12**. It should be noted that each of the logs **15** is not necessarily traveling one after another, or in "single file," but may be partially or fully adjacent to another log **15** on the transitional conveyor platform **32**.

(22) Referring to FIG. 2C, to help urge the logs **15** down the slide **28**, in embodiments, the system may also include a rotating arm **80** having a free, first end **82**, which is round in embodiments, configured to operatively contact the slide **28** (i.e., from direct, or indirect contact, e.g., via a bracket that contacts and supports the slide **28**) on every rotation, thereby shaking or vibrating the slide **28**. In embodiments, as shown in FIG. 2C, a second end **84** of the rotating arm **80** is connected to a rotating axle **86** that also moves the transitional conveyor platform **32** in the direction of travel **24**.

(23) It should be noted that while FIG. 2A shows one complete system **10**, it is contemplated that a second system **10'** may be added adjacent to the system **10** as indicated, that is a mirror image of system **10**, so that a log hopper **14'** of the second system **10'** is side by side with the log hopper **14**

of system **10**, the main conveyor platforms **12**, **12'** of each of the first **10** and second system **10'** run parallel to each other, the slotted conveyor platforms **20**, **20'** extend adjacently in opposite directions in mirror image, and so on, as represented in FIG. 2A. This configuration allows a single truckload of logs to be emptied into both log hoppers **14**, **14'** simultaneously, and also provides double the number of bags of wood to be acquired in the same time it would take the one system **10**.

(24) Referring, for example, to FIGS. 1 and 2A, in embodiments, the system **10** also includes another conveyor platform **34**, which is positioned above and adjacent to a bin portion **39** of the system and is configured to drop the logs **15** into the bin portion **39**. Conveyor platform **34**, which is referred to herein as bin-adjacent platform **34**, intersects the transitional conveyor platform **32**, at its exit end **36**, at a right angle to the direction of travel **24**, so that the longitudinal axis **22** of each of the logs **15** conveyed from the transitional conveyor platform **32** onto the bin-adjacent conveyor platform **34** is now transverse to the direction of travel **24**. Accordingly, once the logs **15** are loaded (could be one at a time or multiple logs **15** at a time) in this transverse orientation onto the bin-adjacent conveyor platform **34**, they are conveyed one after another along the bin-adjacent conveyor platform **34**, with their longitudinal axes **22** oriented parallel to each other. Referring also to FIG. 3, this ensures that the logs **15** will drop one by one, oriented transverse to the direction of travel **24**, off an end **37** of the bin-adjacent conveyor platform **34** and into the bin portion **39**. The bin portion **39** is configured to receive the logs **15** through an opening **33** in its ceiling **38**, and collect the logs **15** in a stack **17**. The longitudinal axis **22** of each of the logs **15** dropped into the bin portion **39** aligns parallel with an inner width **41** of the upper bin portion **39** to maintain a neat stack.

(25) In embodiments, the bin portion **39** includes an upper bin **40** and a lower bin **50**, such that the logs **15** drop off the end **37** of the bin-adjacent conveyor platform **34** and into the upper bin **40**. Advantageously, the transverse direction of travel **24** of the logs **15** on the bin-adjacent conveyor platform **34** ensures that the longitudinal axis **22** of the logs **15** is transverse to the direction of travel **24** as they fall off the end **37** of the bin-adjacent conveyor platform **34** and into the upper bin **40**. This negates the need for any other means of “catching” and aligning each singular log **15** to form the stack **17** of logs **15**. The logs **15** are already oriented in the same direction, with their longitudinal axes **22** parallel to each other, to neatly stack up in the upper bin **40** as they fall off the bin-adjacent conveyor platform **34**.

(26) In embodiments, the system **10** includes a vertical unit **42**, which includes the bin portion **39**, and is, in embodiments, configured to automate the stacking and bagging of logs. Referring to FIG. 3, the vertical unit **42** also preferably includes a first sensor **44**, also referred to herein as a counting sensor **44**, configured to provide a count of the number of logs **15** that pass under it before dropping into the upper bin **40**. The first sensor **44** may be, in embodiments, a photoelectric sensor, for example, a diffuse proximity sensor with the source and detector in a single package, or any other suitable sensor that may be positioned and configured for counting logs **15** that drop into the bin portion **39**. Referring to FIG. 1 as well as FIG. 3, each log is counted as it passes the first (counting) sensor **44**, which, in embodiments, is located above the end **37** of the bin-adjacent conveyor platform **34**, just before one of the logs **15** drops into the upper bin **40**. A bottom floor of the upper bin **40** is formed as a hatch, referred to herein as upper hatch **46**, which is flat in a closed position, and is preferably configured to be hingedly opened and closed around a hinged side **47**. In embodiments, referring also to FIG. 5A, the vertical unit **42** further includes a first (or upper) actuator **60**, which may be a pneumatic actuator, operably connected to the upper hatch **46** and configured to drive the upper hatch **46** open and closed.

(27) Referring also to FIG. 2A, as well as to FIGS. 6A and 6B, in embodiments, a control module **100**, which may be a programmable logic controller, is provided with the system **10**. The system **10** also preferably includes a user control panel **130**, which is in operable communication with the control module **100**, and which allows, inter alia, user selection of control toggles and input of

various parameters. The control module **100** includes a processor **102**, which may be a central processing unit (CPU), a microprocessor, or a plurality of microprocessors, for example, and memory **104**, configured to store parameters, including user-input data, and instructions, or programming steps **108**, which when executed by the processor **102**, performs the steps and processes described herein. The control module **100** is operatively connected to the first sensor **44** and to the upper hatch **46**, and may include a counter **106**, which may be part of the processor **102**, which may be configured to receive a signal from the first (counting) sensor **44** each time a log **15** passes under the sensor **44**, and maintain a running count of the logs **15** based on the number of signals received. In embodiments, a preselected number **111** of logs **15** for forming the stack **17** may be input to the control module **100** by a user via the display screen **128**, such as a touch screen display. In embodiments, the preselected number may be selected via a drop down list of numbers. The preselected number **111** may be displayed on a display screen **128**, which may be a touch screen, in embodiments, associated with the user control panel **130**. The running count of the logs **15** dropped into the bin portion **39** may also be continuously updated and displayed on a counter display portion **132**, also referred to herein as the display counter **132**, of the display screen **128**.

(28) Referring to FIG. 5B, in embodiments, the bin portion **39** also includes a movable wall **64** configured to be translated via linear actuators **66**. Referring also to FIG. 6B, prior to initializing the process of bagging the logs **15**, a user may also enter a length **43** of the logs **15** via the user control panel **130**, which may be displayed on the user control panel **100**. It should be noted that the length **43** of a log **15** is measured along its longitudinal axis **22**, as noted in FIG. 1, and is greater than a diameter or width of the log **15**. After the length **43** is entered, or, in embodiments, selected from a drop down list of lengths displayed on the display screen **128**, the linear actuators **66** will adjust the movable wall **64** so that the inner width **41** of the bin portion **39** corresponds to a width that is based on the length **43** input by the user. For example, the length **43** input by the user may have been adjusted upward by the user to accommodate anomalies in the logs **15** when cut. Alternatively, the control module **100** is configured to set the inner width **41** of the bin portion **39** via the movable wall **64** to the length **43** input by a user plus an additional predetermined percent, so that the inner width **41** is set to accommodate anomalies in the nominal length of the logs **15**.

(29) After the preselected number **111** of logs **15** are counted via the first sensor **44** as having dropped into the upper hatch **46**, the control module **100**, in embodiments, may signal the first actuator **60** to open the upper hatch **46**, dropping all of the logs **15** collected in the upper bin **40** in one stack.

(30) As best seen in FIG. 5A, an additional sensor **61**, referred to as an upper hatch sensor **61**, and which may be, in embodiments, a proximity sensor, such as an inductive proximity sensor, is positioned below a nominal position of the upper hatch **46** when in the closed position and opposite the hinged side **47**. The upper hatch sensor **61** is configured to detect when the upper hatch **46'** is open, the upper hatch **46'** swinging down in front of the upper hatch sensor **61** to trigger a signal to the control module **100**. Likewise, the closing of the upper hatch **46** triggers the upper hatch sensor **61** to send another signal to the control module **100** indicating the upper hatch **46** is closed.

(31) In embodiments, the system may further include a lower bin **50** adjacently below the upper bin **40**, so that when the upper hatch **46** opens, the stack **17** of logs **15** drops into the lower bin **50**. A bottom of the lower bin **50** is also formed as a hatch, referred to herein as a lower hatch **52**, which is preferably configured to be hingedly opened and closed around a hinged side **53**. In embodiments, the lower hatch **52** may be configured to be manually opened via a handle (not shown) attached to the lower hatch **52**, and manually operable to hingedly swing the lower hatch **52** open around its hinged side **53**. In other embodiments, referring to FIG. 5A, the lower hatch **52** may be operable via a second (or lower) actuator **62**, which may be a pneumatic actuator, operably connected to the lower hatch **52** and configured to drive the lower hatch **52** from its closed position to an open position **52'**.

(32) In embodiments, the vertical unit **42** may also include an open portion **56**, having a floor or

platform **55**, also referred to herein as a bagging portion **56**, adjacently beneath the bin portion **39**, e.g., in embodiments, below the lower bin **50**. The open portion **56** is configured for bagging the stack **17** of logs **15** dropped through an opened, lower hatch **52'**. In embodiments, referring also to FIG. **6A**, the lower hatch **52** is configured to open and close via a user operable switch **160**, which is operably connected to, and activates, the second actuator **62** via the user operable switch **160**. Referring also to FIG. **6B**, the user operable switch **160** may be toggled via a foot pedal **162** or a push-button **164**, each of which, when depressed, will open the lower hatch **52** and drop the stack **17** into the bagging portion **56** of the system **10**.

(33) In further embodiments, the system **10** may also include a second (analog) sensor **48**, which may, in embodiments, be an analog sensor, or analog output or displacement sensor. The second sensor **48** is preferably positioned above the bin portion **39**, in embodiments, above the upper bin **40**, and past the end of the bin-adjacent conveyor platform **34**. The second sensor **48** is configured to generate an output signal proportional to a distance **49** (see FIG. **3**, and graphical display of **49'** in FIG. **6B**) between the second sensor **48** and an object in its field of view, namely, an uppermost log **15'** in the upper bin **40**, or when the upper bin **40** is empty, the upper hatch **46**. Accordingly, the output signal can be processed by the control module **100** to determine whether the upper bin **40** is empty, and also to keep updating the distance **49**. Referring also to FIGS. **6A** and **6B**, in embodiments, each of the counter **106** in the control module **100** and the display counter **132** in the display screen **128**, is reset to zero to begin the counting process via the first sensor **44** again in response to receiving a signal from the second sensor **48** that the upper bin **40** is empty.

(34) In other embodiments, the display counter **132** and counter **106** are reset to zero in response to the control module **100** receiving a signal from either the first (upper) actuator **62**, or, in embodiments, from the upper hatch sensor **61**, that the upper hatch **46** has opened and closed. In embodiments, once some additional checks have been performed, such as rechecking via the second sensor **48** that the upper bin **40** is empty, the upper bin **40** may begin to be filled up again. While the upper bin **40** begins to fill again, the lower hatch **52** may be opened via the user operable switch **160**, which activates the second actuator **62**, to drop the stack **17** of logs **15** into a bag **54**, for example, positioned in the bagging portion **56** of the vertical unit **42** located below the lower bin **50** of the bin portion **39** and lower hatch **52**.

(35) In further embodiments, referring also to FIG. **6B**, to account for variations in the sizes of the logs **15**, and to form stacks **17** of logs **15** of similar volume or height, the distance **49** of the uppermost log **15'** from the sensor **48** is also monitored and compared to a predetermined distance **126**, which may be stored in the memory **104** of the control module **100**. The predetermined distance **126** may be input by a user, or optionally, a predetermined parameter **127**, such as a height, volume, or weight of a stack **17** of logs **15** to be collected in the upper bin **40** before opening the upper hatch **46**, may be input by a user via the user control panel **100** from which the predetermined distance **126** may be calculated via the processor **102**, based on a height of the upper bin **40** and, in embodiments, on the length **43** of the logs **15** input by the user. In embodiments, while the counter **106** still monitors the number of logs **15** dropped into the upper bin **40**, the control module **100** continues to allow logs **15** to drop into the upper bin **40** until the distance **49** of the uppermost log **15'** in a stack **17** to the second sensor **48** reaches, i.e., equals or exceeds than (by one log) the predetermined distance **126** from the second sensor **48**. In further embodiments, the control module **100** signals the first actuator **60** to open the upper hatch **46**, dropping all of the logs **15** collected in the upper bin **40** in one stack, when this condition is achieved, i.e., in response to the control module **100** receiving a signal indicating that the distance **49** equals or exceeds the predetermined distance.

(36) Referring to FIG. **6B**, in embodiments, the user control panel **130** may include a graphical indicator **134**, e.g., a data bar display, which is configured to graphically portray, in embodiments, as a colored bar **135**, the rising height of the stack **17** in the upper bin **40**, based on the distance **49**.

(37) Referring to FIGS. **3** and **4**, the vertical unit **42**, in embodiments, may include the bagging

portion **56**, which preferably includes the platform **55** on which a bag **54** filled with the stack **17** of logs **15** may rest, and may be configured to accommodate any type of bags, including, but not limited to, mesh, and plastic. In embodiments, a bag support flap **70** may be positioned below the lower hatch **52** with a front support bar **74** positioned in front of it, which hingedly rests against a back of a bag, and preferably, which holds a number of bags. A front of a single bag **54** may be pulled around a frame **76** positioned below the lower hatch **52** for bagging, wherein the single bag **54** may be removed after filling the bag **54** with logs **15** dropped through the lower hatch **52**.

(38) In additional embodiments, as shown in FIG. 3, for example, the system **10** may also include a third sensor **58**, which may be a photoelectric sensor, for example, a diffuse proximity sensor, which is positioned and configured to determine whether the lower bin **50** is full or empty. In embodiments, the third sensor **58** may be located adjacent and just above the lower hatch **52** of the lower bin **50**. As one of skill in the art will appreciate, the third (proximity) sensor **58** may be configured to switch between an on and off state in the presence and absence, respectively, or vice versa, of light reflected off a log **15** and back into the third sensor **58**. Accordingly, the control module **100** may be configured, in embodiments, to activate the lower actuator **62** to automatically open the lower hatch **52** and release the stack **17** from the lower bin **50**, in response to a signal from the third sensor **58** indicating the presence of logs **15** in the lower bin **50**, while the upper loading bin **40** is being refilled. This can significantly increase the bagging process compared to prior art systems.

(39) Referring to FIG. 6B, a switch **158** may be provided on the user control panel **130** that allows the operation of the lower hatch **52** to be set to manual operation, so that it is opened and closed via the manually operable switch **160**, or to automated operation, so that it is automatically opened and closed via the control module **100**.

(40) The control module **100** may also be configured to identify, via the third sensor **58**, if there are any logs **15** remaining after the lower actuator **62** has opened and closed, indicating that the lower bin **50** needs to be cleared before refilling the lower bin **50** with another dump of the stack of logs **15** from the upper bin **40**. The display screen **128** associated with the control module **100** includes, in embodiments, a momentary switch **142** operable by a user, in both the manual operation and automated operation setting of the switch **158**, which is operatively connected to the lower actuator **62**, and is configured to open the lower hatch **52** via lower actuator **62** for as long as the momentary switch **142** is engaged by the user, to clear the lower bin **50**. In embodiments, the processor **102** is configured to stop the conveyor system, via the motor controller **35**, as well as to move the linear actuators **66** to increase the inner width **41** to its maximum width, and open the lower hatch **52** in response to the momentary switch **142** being activated by the user. In embodiments, the processor **102** is configured to also open the upper hatch **46**, via upper actuator **60**, when the momentary switch **142** is activated by the user to dislodge any stuck logs in the bin portion **39** and clear both the upper **40** and lower bin portion **50** from all logs before beginning to fill the upper bin **40** again.

(41) Referring to FIG. 6A, the control module **100** includes memory **104**, which stores a program **108**, i.e., a set of instructions or programming steps **108**, that when executed by the processor **102**, performs the steps of the various methods and processes described herein. Data needed in the control of the system may also be stored in the memory **104**, which may include non-volatile, e.g., flash memory, and/or volatile memory, e.g., random access memory (RAM).

(42) Still referring to FIG. 6A, for communicating with the various sensors, actuators, drivers, motors and motor speed controllers, for example, of the system **10**, and, in embodiments, with an external device **118**, which may be any type of computer, e.g., the control module **100** also preferably includes a communication interface **110**. The external device **118**, in embodiments, may be used to upload updates to the program **108** and system parameters, and to log data generated by the bagging system **10** related to operation and efficiency in the bundling and bagging process.

(43) The control module **100** may also include, or be in operative communication with, an input module **112** and an output module **114** configured to receive and send data and/or signals from

input devices and to output devices, respectively. For example, the processor **102**, in embodiments, receives input signals **115** from the sensors via the input module **112**, including counting sensor **44**, analog sensor **48**, lower bin sensor **58**, and upper hatch sensor **61**, and executes the program instructions **108** that are stored in memory **104** based on the input signals **115** and/or data from the sensors. The processor **102** then sends output signals **116** and/or data based on the program instructions **108** and input signals **115** to the output devices, in accordance with the methods of the present disclosure, e.g., to actuators, such as upper actuator **60**, lower actuator **62**, and wall actuators **66**, and to the user control panel **130**.

(44) The processor **102** is also configured to receive input signals **115** and data from the user control panel **130**, in embodiments, via the input module **112**, based on user input to the control panel **130**, via the display screen **128** operatively connected to the user control panel **130**, and on input signals **115** and data from the sensors. The display screen **128**, which may be a touch screen display, is configured for both user input and display of data. The user control panel **130** may also be configured to send output signals or data **116** to the user control panel **130** for display on the user control panel **130**, based on user input and/or on input signals **115**, as further described herein. The processor **102** is further configured to execute instructions **108** via the output module **114** to certain output devices, as further described herein, based on user input to the user control panel **130**.

(45) Referring to FIG. 2A, as well as FIG. 6A, the power to, and speed of, all of the conveyor platforms, i.e., main conveyor platform **12**, slotted conveyor platform **20**, transitional conveyor platform **32**, and the bin-adjacent conveyor platform **34** may be powered by a motor **45**, e.g., an electric motor, which may be controlled by a motor speed controller **35**. A speed **136** of the conveyor system may be manually input via a user input device **144** on the user control panel **130**, which may include keys, which may be touch screen keys, or a dial, which may be a touch screen dial, associated with the control module **100**, and may be displayed on the display screen **128**. The processor **102** then sends the speed **136** data, via output module **114**, for example, to the motor speed controller **35** to drive the conveyor system.

(46) The processor **102** also executes instructions, via output module **114**, to the conveyors, to automatically stop and start, based on input signals from the sensors, as further described herein.

(47) It should be recognized that while the control module **100** may, in embodiments, be packaged as a single unit together with the processor **102**, memory **104**, and communication interface **110**, as shown, the term control module **100** should not be construed as requiring these components to be packaged together. In embodiments, one or more of the components may be operatively connected to the processor **102** and memory **104**, but not co-located therewith. Likewise input modules **112** and output module **114** may be packaged separately from the processor **102**, memory **104**, and communication interface **110**, or may also be part of a “packaged” unit, for example, a programming logic controller.

(48) It should also be appreciated that the input module **112** and output module **114** of FIG. 6A may represent a plurality of input and output modules, respectively, each of which is configured for digital input and output, or analog input and output, respectively. For example, input modules will be configured based on the type of sensor, and on the control panel input. Likewise, some of the output modules **114** may be configured for digital output, for example, to on/off switches **68,69**, and others may be configured for analog output, for example, to adjust the speed of the conveyors in response to analog user input via the user control panel **130**.

(49) Referring now to FIG. 6B, the user control panel **130** includes, in embodiments, the display screen **128**, which may be a touch screen display. The user control panel **130** may also include a user-operable on switch **138**, which may be, for example, a (start) push button **138**, a user-operable off switch **140**, which may be, for example, a (stop) push button **140**, the (bin clear) momentary switch **142**, and, in embodiments, an emergency stop switch **143** configured to turn off all power to the system **10**. As described herein, the user control panel **130** may also be configured to accept

user inputs via the display screen **128** for various parameters, which may be processed by the processor **102** and/or displayed on the display screen **128**, including conveyor speed **136**, log length **43**, and a preselected number of logs **111**.

(50) The user control panel **130** is configured to send this user input to the control module **100**, which handles inputs and outputs to the various actuators, solenoids, motors and sensors, as described herein, for example, via the input modules **112** and output modules **114**. The control module **100** is programmed to monitor sensor data and user inputs to perform the automated decisions to complete the bagging process, in accordance with the methods of the present disclosure.

(51) Additional data displayed on the user control panel **130** may include a bag counter **146** that provides a running count of the bags completed during a start cycle. The bag counter **146** is updated by one every time the lower hatch **52** opens and closes in response to activation of the user operable switch **160** before a cycle stops, e.g., via activation of the stop push button **140** on the user control panel **130**.

(52) The display screen **128** may also display the status of various operating parameters of the system **10**, configured for easy monitoring by the user. For example, in embodiments, the display screen **128** may include a conveyor status indicator **148** of the conveyor system, including the bin-adjacent conveyor platform **34**, to indicate that the conveyor system is on, in response to the user-operable on switch **138** being activated, and after verifying the system **10** is ready to operate without any errors, or off, in response to the user-operable off switch **140** being activated. The indicator **148** may be provided, in embodiments, as words, pictorial graphics, and/or colors, e.g., green for on, and red for off.

(53) In additional embodiments, a graphical indicator **134**, in embodiments, a data bar display, may also be displayed on the display screen **128**, which is configured to graphically display, as a colored bar **135**, the volume of logs **15** stacked in the upper bin **40**. For example, the colored bar **135** may increase in height with every log **15** dropped into the bin, such that a distance **49'** between the top of the colored bar and the top of the data bar display represents the distance **49** (see FIG. 3) between the second sensor **48** and the uppermost log **15'** in the upper bin **40**.

(54) A lower bin status indicator **150** may also be provided on the display screen **128**, to indicate whether the lower bin **50** is empty, or filled with the stack **17** of logs **15** dropped through the lower hatch **52**, or still occupied with at least one log **15** after the lower hatch **52** is actuated, based on the signal from the third (proximity) sensor **58**, indicating that the lower bin **50** is empty or occupied, after the lower hatch **52** has opened and closed, in accordance with the methods of the present disclosure. Any suitable type of indicator may be used alone, or in combination, such as text, color, bar display, or graphics.

(55) Referring to FIG. 7, an embodiment of a method **170** of the present disclosure may include initiating a bagging cycle, at **172**, in response, for example, to activating, at **174**, via a user-activated on switch **138**, e.g., a start push button, on the user control panel **130**.

(56) Initiating the bagging cycle, at **172**, may further include the control module **100**, via the processor **102**, checking input signals **115** from a plurality of sensors **80** (see FIG. 6A) associated with the bin portion **39** of the system to ensure that the bin portion **39**, which may include upper bin **40** and lower bin **50**, are empty and that the upper hatch **46** is closed and ready to receive logs **15**. For example, in embodiments, the initiating step includes checking the second sensor **48**, which is positioned above the upper bin and past the end of the bin-adjacent conveyor **34**, to ensure that the upper bin **40** is empty, checking the third sensor **58** to ensure that the lower bin **50** is empty, and checking via the control module **100**, the input signal **115** from the upper hatch sensor **61** to ensure the upper hatch **46** is closed.

(57) In embodiments, the method includes starting the conveyor system, at **182**, including the bin-adjacent conveyor platform **34**, at the completion of the initiating cycle, at **172**, via the processor **102** executing instructions to the motor controller **35**. In embodiments, the conveyor system is

started in response to a user activating the on switch **138**, at **174**. Logs **15** begin traveling on the conveyor system, eventually being conveyed, at **178**, on the bin-adjacent conveyor platform **34** and toward the bin portion **39**, with the logs **15** oriented transversely to the direction of travel **24** of the bin-adjacent conveyor platform **34**, in embodiments, so that the logs dropped sequentially off the end of the bin-adjacent conveyor platform are received into the bin portion **39**, at **180**, in embodiments, into the upper bin **40**, to form a stack.

(58) Embodiments of the method **170** may further include the control module **100**, via the processor **102**, sending an updated status of the conveyor system, at **182**, to the user control panel **130**, which is displayed in the status indicator **148** portion of the display screen **128**.

(59) In embodiments, the method may further include detecting each log, at **184**, via the first sensor **44**, which is positioned before the end of the bin-adjacent conveyor platform **34**, each log as it passes the first sensor and drops into the bin portion **39**. In embodiments, the control module **100** continuously monitors the input signal **115** from the first (counting) sensor **44**, the method further including maintaining, at **186**, via the processor **102**, a running count of the logs **15** that drop into the upper bin **40**, based on the number of logs **15** that pass the first sensor **15**. The method may further include, as part of the updating status steps, at **182**, for example, updating the display counter **132** on the display screen **128**, via the control module **100**, each time a log **15** passes the first sensor **44**, to maintain the running count of the logs **15**. The method further includes continuing the maintaining the running count **186** and the conveying logs steps until the running count equals the preselected number **111** of logs.

(60) In embodiments, the method may further include measuring a distance, at **188**, via the second sensor **48**, from the second sensor **48** to an uppermost log **15'** of the stack of logs in the upper bin **40**, or to the upper hatch **46** in an absence of logs in the upper bin **40**.

(61) For example, in embodiments, the control module **100** may continuously check the input signal **115** from the second sensor **48**, which is configured to generate an output signal proportional to a distance **49** from the second sensor **48** to the uppermost log **15** of the stack **17** in the upper bin **40**, or from the upper hatch **46** when the upper bin **40** is empty, the method further including, in embodiments, communicating the distance **49**, at **190**, via the second sensor **48**, to the control module **100**.

(62) In further embodiments, at **182**, for example, the control module **100** may also update the graphical indicator **135** on the display screen **128**, wherein the graphical indicator is configured to graphically portray the distance **49** decreasing, i.e., the stack **17** rising in the upper bin **40**.

(63) In response to either the running count of logs **15** equaling the preselected number **111** of logs **15** or the distance from the second sensor **48** and the uppermost log **15'** reaching a predetermined distance **126**, the method further includes automatically opening the upper hatch **46**, at **192**, via the upper actuator **60** in communication with the control module **100**. In embodiments, for example, once the running count of the logs **15** detected, at **184**, equals the preselected number, the distance **49** is compared, via the processor **102**, to the predetermined distance **126**. If the distance **49** is less than the predetermined distance **126**, the bin-adjacent conveyor platform **34** continues to drop logs **15** into the upper bin **40** until the distance **49** equals, or is less than the predetermined distance. The upper hatch **46** opens in response to the distance **49** equaling, or exceeding the predetermined distance.

(64) In further embodiments, the method further includes stopping, at **194**, the bin-adjacent conveyor **34**, via the motor controller **35** in communication with the control module **100**, prior to actuating the upper hatch **46** to open, at **192**, in response in response to one of the running count of logs reaching a preselected number of logs, and the distance from the second sensor and the uppermost log reaching the predetermined distance **126**, thereby dropping the logs **15** in one stack **17** into the lower bin **50**.

(65) Consequently, the third sensor **58**, which is positioned and configured to detect a presence of logs **15** in the lower bin **50**, for example, adjacent to the lower bin **50** just above the lower hatch **52**,

is now blocked. The method may further include detecting the stack of logs **15**, at **198**, in the lower bin **50**, via the third sensor **58**. In embodiments, the control module **100** updates, at **182**, for example, the lower bin status indicator **150** to indicate it is full, and the graphical indicator **134** for the upper bin **40**, to indicate it is empty.

(66) In response to receiving the signal from the third sensor **58** indicating that the stack **17** is now received in the lower bin **40**, the upper hatch **46** is actuated closed, via the first, upper, actuator **60**, and the bin-adjacent conveyor **34** is restarted, at **202**, in accordance with instructions received from the processor **102**.

(67) In embodiments, restarting the bin-adjacent conveyor **34**, at **202**, resets the running count based on the first signal from the first sensor **44**, to zero. In embodiments, the method further includes resetting, via the control module **100**, the display counter **132** to zero, based on the updated running count, and, in further embodiments, sending updated status data, at **182**, for example, to the lower bin status indicator **150** on the display screen **128**, indicating the lower bin **50** is full and ready for emptying, for example, into the open portion **56** below the lower bin **50** for bagging. In embodiments, the control module **100** also sends updated data, at **182**, to the graphical indicator **134** on the display screen **128** to display the updated empty status of the upper bin **40**.

(68) In embodiments, a user seeing that the lower bin **50** is full may activate the manually operable switch **160** to open and close the lower the lower hatch **52**, at **204**, via the lower actuator **62**, thereby emptying the lower bin **50**, preferably while upper bin **40** is already refilling, and in embodiments, after restarting the bin-adjacent conveyor platform **34**, at **202**.

(69) In embodiments, in response to the activation and deactivation of the manually operable switch **160**, a number of bags displayed in the bag counter **146** portion on the display screen **128** is updated by one, via the control module, at **182**, for example. The bag counter **146** continues to be updated with every opening of the lower hatch **52** until the run cycle is completed, e.g., until the off switch **69** is enabled, for example, via the stop push button or other user-activated off device **140** on the user control panel **130**.

(70) While particular embodiments of the present invention have been particularly shown and described with reference to specific embodiments, it should be apparent to those skilled in the art that the foregoing is illustrative only and not limiting, having been presented by way of example only. It is to be understood that the disclosed embodiments are merely examples, which may be embodied in various forms and detail without departing from the spirit and scope of the invention. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting. Numerous other embodiments may fall within the scope of the accompanying claims and equivalents thereto.

Claims

1. A system for bundling wood, the system including comprising: a bin-adjacent conveyor platform configured to transport logs in a transverse orientation relative to a direction of travel of the bin-adjacent conveyor platform, a longitudinal axis of each of the logs on the bin-adjacent conveyor platform being in a transverse orientation relative to the direction of travel; and a bin portion, wherein an end of the bin-adjacent conveyor platform is positioned above the bin portion, such that each log sequentially drops off the end of the bin-adjacent conveyor platform in the transverse orientation into the bin portion; a first sensor positioned before the end of the bin-adjacent conveyor platform, the first sensor configured to detect each log as the log passes the first sensor in the transverse orientation and before the log sequentially drops off the end of the bin-adjacent conveyor platform; a lower hatch formed in a bottom of the bin portion; and a control module, operatively connected to the first sensor, the control module including a processor, and configured to maintain a running count of logs detected by the first sensor; wherein the lower hatch is configured to hingedly open and simultaneously drop a stack of logs therethrough based on the

running count.

2. The system of claim 1, wherein the bin portion includes an upper bin including an upper hatch formed in a bottom of the upper bin and an upper actuator operatively connected to the control module and to the upper hatch, and a lower bin positioned below the upper bin, the lower bin including the lower hatch, wherein the upper bin is configured to receive each log detected by the first sensor and sequentially dropped off the end of the bin-adjacent conveyor platform to form the stack of logs, and wherein the control module is operatively connected to the upper actuator and is configured, via the upper actuator, to automatically open the upper hatch to drop the stack of logs into the lower bin based on the running count of logs.

3. The system of claim 2, wherein the control module is configured, via the upper actuator, to automatically open the upper hatch to drop the stack of logs into the lower bin in response to the running count of logs equaling a preselected number of logs.

4. The system of claim 2, further comprising a second sensor positioned above the bin portion and past the end of the bin-adjacent conveyor platform, wherein the second sensor is operatively connected to and in communication with the control module and is configured to measure a distance from the second sensor to an uppermost log of the stack of logs in the upper bin, or to the upper hatch in an absence of logs in the upper bin, and to communicate the distance to the control module, wherein the control module is configured, via the upper actuator, to automatically open the upper hatch to drop the stack of logs into the lower bin in response to one of the running count of logs reaching a preselected number of logs and the distance from the second sensor and the uppermost log reaching a predetermined distance.

5. The system of claim 4, further including a third sensor positioned and configured to detect a presence of logs on the lower hatch in the lower bin, the third sensor being operatively connected to and in communication with the control module, wherein the control module is configured, via the upper actuator, to automatically close the upper hatch in response to the third sensor detecting the logs in the lower bin corresponding to the stack of logs dropped into the lower bin from the upper bin.

6. The system of claim 5, wherein the control module is further configured to reset the running count to zero in response to the automatic closing of the upper hatch.

7. The system of claim 4, wherein the control module is further configured to stop the bin-adjacent conveyor platform and then automatically open the upper hatch to drop the stack of logs in response to one of the running count of logs reaching the preselected number of logs and the distance from the second sensor and the uppermost log reaching a predetermined distance.

8. The system of claim 7, wherein the control module is further configured to restart the bin-adjacent conveyor platform in response to the automatic closing of the upper hatch.

9. The system of claim 6, further including a user control panel operatively connected to the control module and a display screen operatively connected to and associated therewith, the display screen including at least one of a graphical indicator of a status of the upper bin and a display counter that continuously displays a number of logs in the upper bin based on the running count of logs, wherein the control module is configured to continuously update the at least one of the graphical indicator and the display counter to display the number of logs collected in the upper bin and to indicate, in response to the automatic closing of the upper hatch, an empty status of the upper bin.

10. The system of claim 5, further including a user control panel operatively connected to the control module and a display screen operatively connected to and associated therewith, the display screen including a lower bin status indicator configured to display a status of the lower bin as empty or full based on the third sensor, wherein the status switches to full in response to the third sensor detecting the logs in the lower bin corresponding to the stack of logs dropped into the lower bin from the upper bin.

11. The system of claim 1, further including a user control panel operatively connected to the control module, and a bagging portion adjacently below the lower bin, the user control panel

including a user operable switch operatively connected to a lower actuator and configured to allow a user to open and close the lower hatch in use, via the user operable switch, to drop the stack of logs therethrough and into the bagging portion.

12. The system of claim 5, further including a lower actuator operatively connected to the control module and to the lower hatch, and a bagging portion positioned adjacently below the lower bin, and wherein the control module is configured to automatically open the lower hatch, via the lower actuator, and release the stack of logs into the bagging portion in response to the third sensor detecting the presence of logs in the lower bin corresponding to the stack of logs dropped from the upper bin.

13. The system of claim 1, further including: a main conveyor device onto which the logs are fed in a random orientation from a log hopper; a slotted conveyor platform positioned to transport the logs that exit from the main conveyor device, wherein the slotted conveyor platform includes a plurality of flat plates spaced apart on each of a plurality of rotatable axles, the plurality of rotatable axles being parallel to each other and spaced apart along the direction of travel of the slotted conveyor platform, wherein the plurality of rotatable axes and the plurality of flat plates are positioned and configured to urge each of the logs in a longitudinal orientation with the longitudinal axis aligned with the direction of travel of the slotted conveyor platform; and a transitional conveyor belt positioned to move and transfer the logs that exit the slotted conveyor platform to the bin-adjacent conveyor platform in the longitudinal orientation, wherein the bin-adjacent conveyor platform intersects the transitional conveyor belt perpendicular to the direction of travel of the transitional conveyor belt, such that the logs exiting the transitional conveyor belt are oriented in the transverse orientation relative to the direction of travel of the bin-adjacent conveyor platform.

14. The system of claim 13, wherein the slotted conveyor platform is positioned at a height above an input end of the transitional conveyor belt, the system further including a slide operatively connected between an exit end of the slotted conveyor platform and the input end of the transitional conveyor belt, wherein the slide is concave in cross-section transverse to the direction of travel, and wherein the direction of travel of the slide is curved and oriented to direct the logs in a downward and curved path and onto the input end of the transitional conveyor belt.

15. The system of claim 13, wherein the main conveyor device is positioned parallel to the transitional conveyor belt, with the direction of travel of the transitional conveyor belt being parallel to but opposite that of the main conveyor device.

16. The system of claim 14, further including a rotating arm connected to an outward extension of one axle of the plurality of rotatable axles proximate the slide and configured to operatively strike a portion of the slide as the rotating arm rotates with the one axle, the rotating arm configured to shake the slide to urge the logs downward along the slide and to agitate the logs to loosen debris therefrom, wherein the plurality of flat plates is also configured to agitate the logs urged along the direction of travel.

17. A method of bundling wood for bagging, the method comprising: receiving into an upper bin, logs dropped sequentially off an end of a bin-adjacent conveyor platform to form a stack of logs in the upper bin, the upper bin including an upper hatch formed in a bottom floor of the upper bin; providing a first sensor positioned before the end of the bin-adjacent conveyor platform, a second sensor positioned above the upper bin, an upper actuator configured to open and close the upper hatch, and a control module operatively connected to, and in communication with, the first sensor, the second sensor, the upper actuator, and the bin-adjacent conveyor platform; detecting each log, via the first sensor, as the log passes the first sensor and before the log sequentially drops off the end of the bin-adjacent conveyor platform; maintaining, via the control module, a running count of logs detected, via the first sensor, and dropped into the upper bin; measuring a distance, via the second sensor, from the second sensor to an uppermost log of the stack of logs in the upper bin, or to the upper hatch in an absence of logs in the upper bin; communicating the distance, via the

second sensor, to the control module; and automatically opening the upper hatch, via the upper actuator in communication with the control module, in response to one of the running count of logs reaching a preselected number of logs, and the distance from the second sensor and the uppermost log reaching a predetermined distance, thereby dropping the stack of logs into a lower bin.

18. The method of claim 17, further including stopping, via the control module, the bin-adjacent conveyor platform, in response to the one of the running count of logs reaching a preselected number of logs, and the distance from the second sensor and the uppermost log reaching a predetermined distance, and prior to the automatically opening the upper hatch.

19. The method of claim 18, further including: providing a third sensor positioned and configured to detect a presence of logs in the lower bin; detecting the stack of logs, via the third sensor, in the lower bin; automatically closing, via the upper actuator, the upper hatch in response to the detecting the stack of logs; and restarting the bin-adjacent conveyor platform, via the control module, and resetting the running count to zero, via the control module, in response to the automatically closing the upper hatch.

20. The method of claim 19, further including: providing a lower hatch formed in a bottom of the lower bin, and a lower actuator configured to open and close the lower hatch; and opening and closing the lower hatch, via the lower actuator, to empty the lower bin, after the restarting the bin-adjacent conveyor platform step.

21. The method of claim 17, wherein the receiving into the upper bin step includes receiving each log dropped sequentially off the end of the bin-adjacent conveyor platform in a transverse orientation, a longitudinal axis of each log being transverse to a direction of travel of the bin-adjacent conveyor platform.

22. A vertical unit for bundling a stack of wood received from a conveyor platform, the vertical unit comprising: an upper bin positionable to receive logs dropped from a conveyor platform, the upper bin including an upper hatch formed in a bottom floor of the upper bin; a lower bin positioned below the upper bin, the lower bin including a lower hatch formed in a bottom floor of the lower bin; a first sensor positionable above an end of the conveyor platform, the first sensor configured to detect each log as the log passes the first sensor; a control module operatively connected to the first sensor, the control module including a processor, and configured to maintain a running count of logs detected by the first sensor; an upper actuator operatively connected to the control module and to the upper hatch; a second sensor positioned above a bin portion, wherein the second sensor is operatively connected to and in communication with the control module and is configured to measure a distance from the second sensor to an uppermost log of the stack of logs in the upper bin, or to the upper hatch in an absence of logs in the upper bin, and to communicate the distance to the control module, wherein the control module is configured, via the control module, to automatically open the upper hatch, via the upper actuator, to drop the stack of logs into the lower bin in response to one of the running count of logs reaching a preselected number of logs and the distance from the second sensor and the uppermost log reaching a predetermined distance; and a third sensor positioned and configured to detect a presence of logs on the lower hatch in the lower bin, the third sensor being operatively connected to and in communication with the control module, wherein the control module is configured, via the upper actuator, to automatically close the upper hatch in response to the third sensor detecting the stack of logs dropped into the lower bin from the upper bin; wherein the control module is further configured to reset the running count to zero and to restart the conveyor platform in response to the automatic closing of the upper hatch.
